



Playful Minds

Parental playfulness within the relation of pretend play and executive functioning



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UCD Babylab, August 27th, 2026

What is Play?

"Play is a very 'fuzzy' concept."

(Lillard 2013, p. 426)



Jean Piaget



Lev Vygotsky



Maria Montessori



What is Play?

“When you’re having fun doing what you’re doing and you like to do it.”

(Lucy, cited in Miller et al. 2008, p. 410)



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What is Play?

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Continuum of Play

Not Playful



Fully Playful



What is Play?

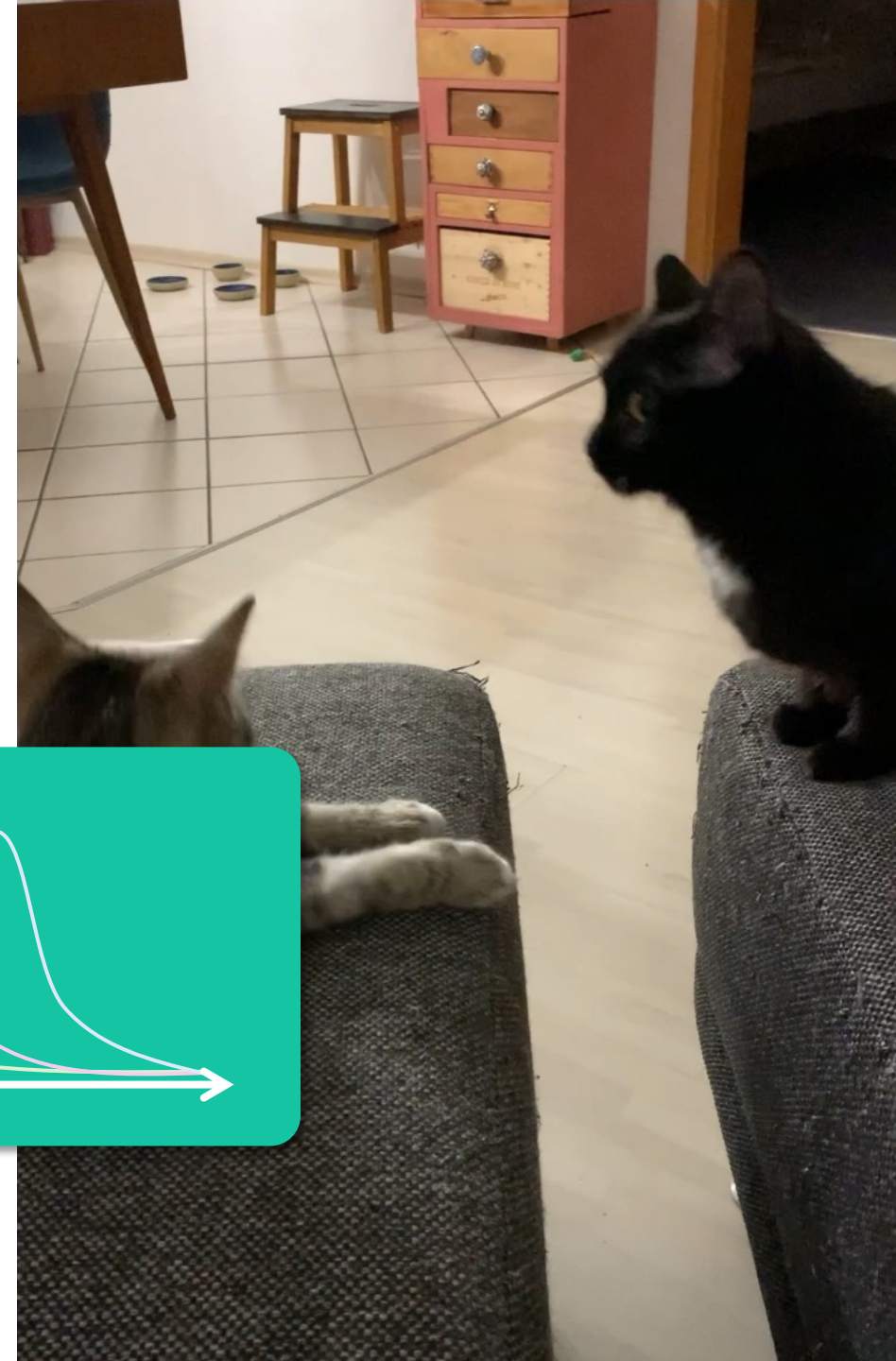
“When you’re having fun doing what you’re doing and you like to do it.”

(Lucy, cited in Miller et al. 2008, p. 410)

Varies categories of play exist.

Sensorimotor play, physical play, rough-and-tumble play, etc.

Prevalence varies across age.



Pretend Play

- Involves imagination, creativity, and fantasy.
 - Object substitution
 - Object attribution
 - Role play
- Emerges around 18 months of age and peaks during the preschool years.
- Related to the development of social competence, language, and executive functions.

Constien et al., 2026; Li & Zhang, 2026; Quinn et al., 2018; Smits-van der Nat et al., 2024





Linking Pretend Play and EF

**Executive
Functions**

Representational
Competence

**Pretend
Play**





Adults play too!



Relationship
satisfaction



Physical
Activity



Mental
Health



Parenting

The Current Study

Hypothesis 1

Toddlers' ability to engage in pretend play will positively predict toddlers' executive functions.

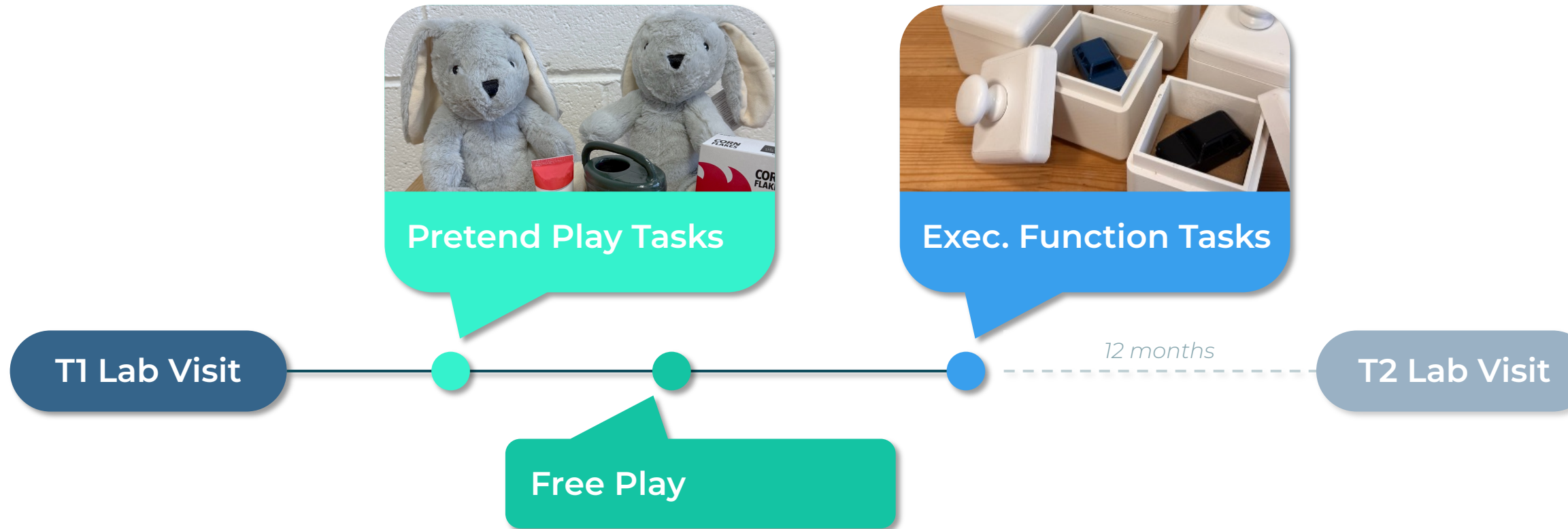
Hypothesis 2

Parental playfulness will be positively associated with toddlers' ability to engage in pretend play.

pre-registered on OSF



The Current Study





Participants

● 100 Toddlers

- $M = 23.51$ months ($SD = 2.75$)
- Range: 20 – 28 months



- 86% White Irish, 8% other White, 6% other ethnic background

● 100 Parents

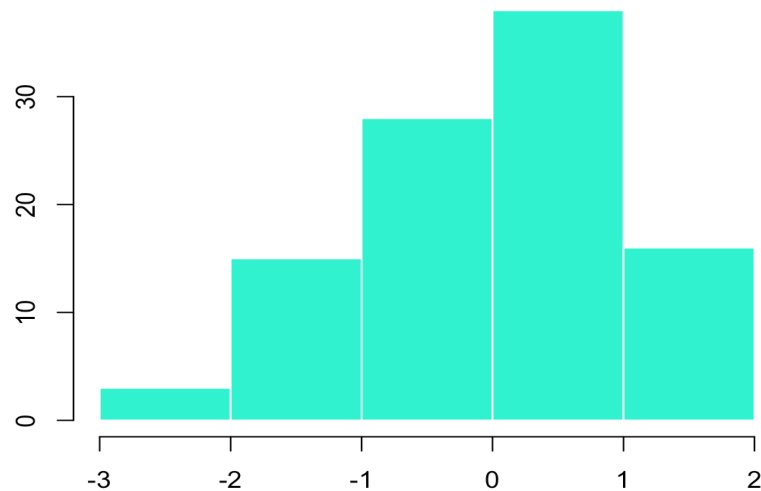
- 90% Mothers, 10% Fathers
- 93% third-level education

Measures

● Pretend Play

- Aggregated Score across three tasks:

Pretend Play Comprehension | Elicited Pretend Play Actions | Impersonation Interview



Standardised Score

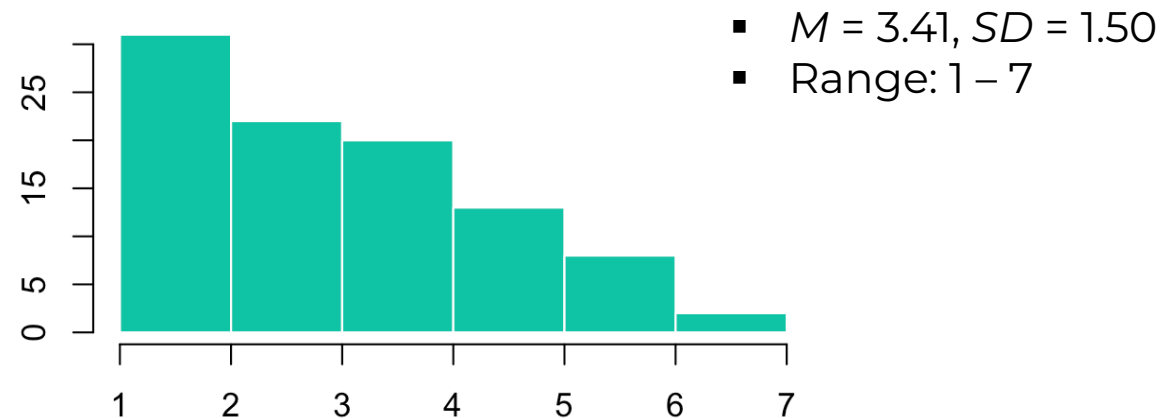
- $M = 0, SD = 1$
- Range: -2.28–1.66



Measures

● Parental Playfulness

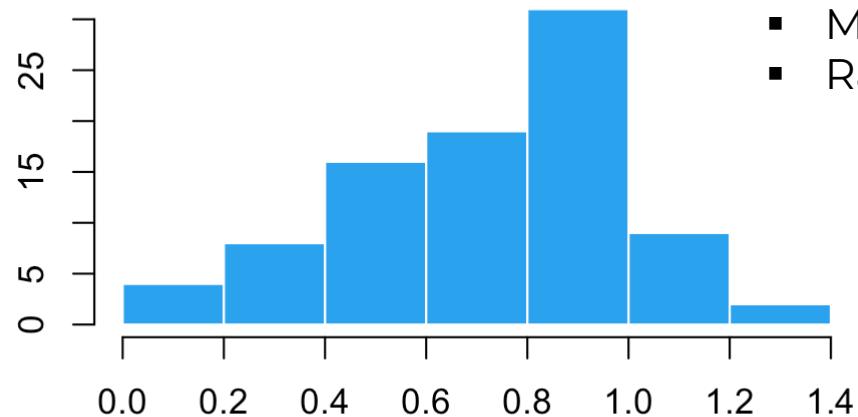
- The extent to which a parent shows **imagination, creativity, and enjoyment** in play with their child.
- Global score using an observational coding scheme.



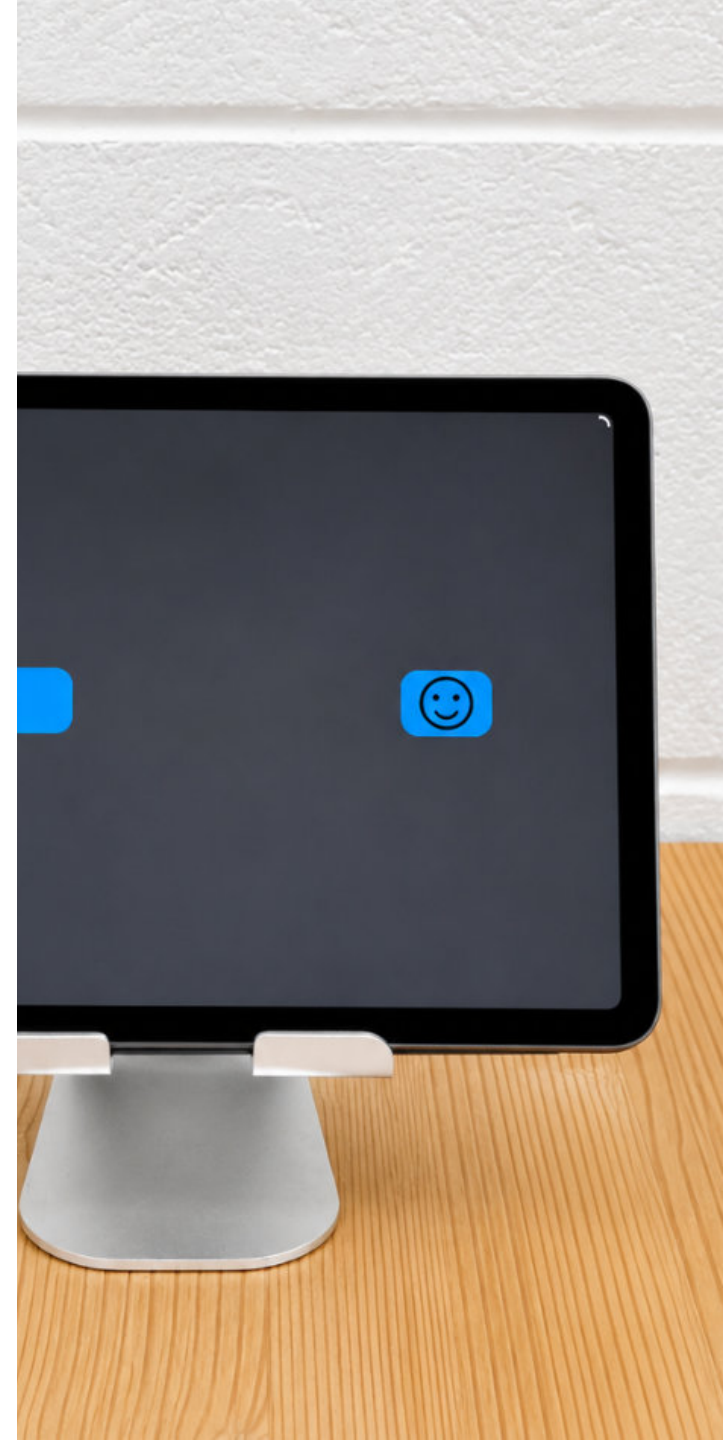
Measures

● Early Childhood Inhibitory Touchscreen Task

- Used to assess response inhibitory control.
- Accuracy score on inhibitory trials (adjusted for prepotent accuracy)



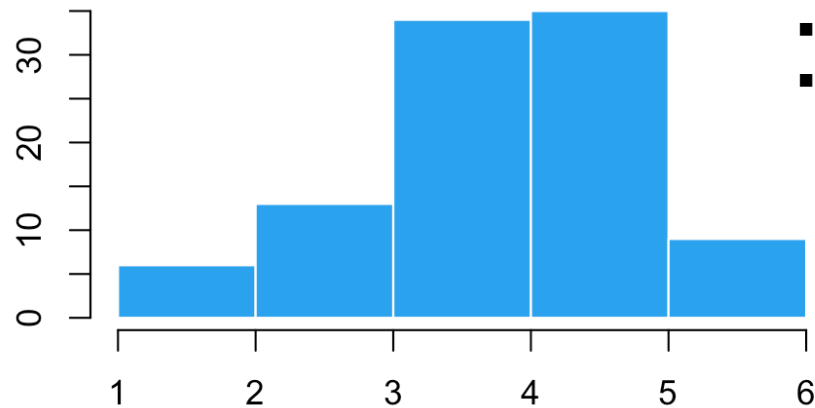
- $M = 0.74$, $SD = 0.28$
- Range: 0 – 1.33



Measures

● Six Boxes Task

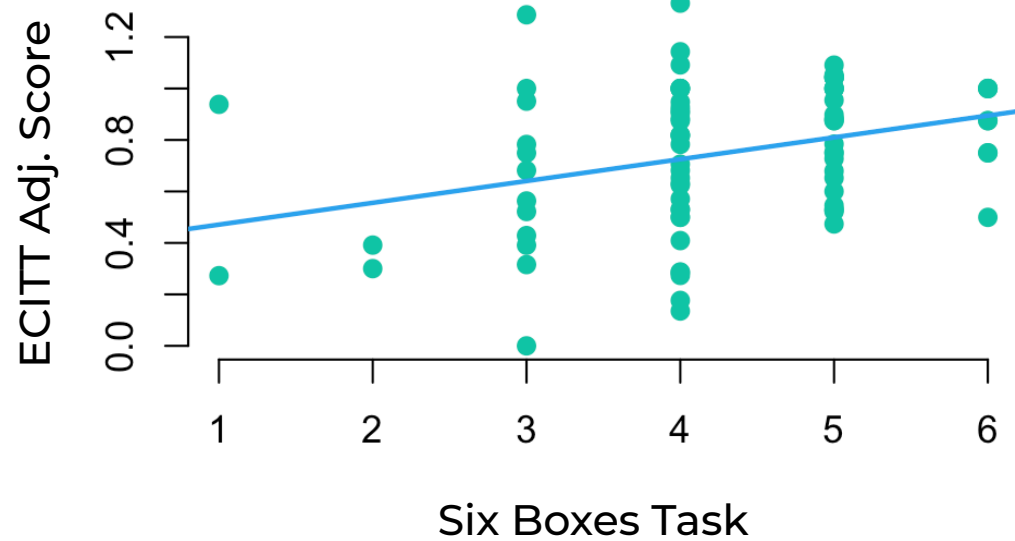
- Used to assess working memory.
- Scored by number of retrieved cars after six trials.



- $M = 4.26, SD = 1.1$
- Range: 1 – 6 Cars



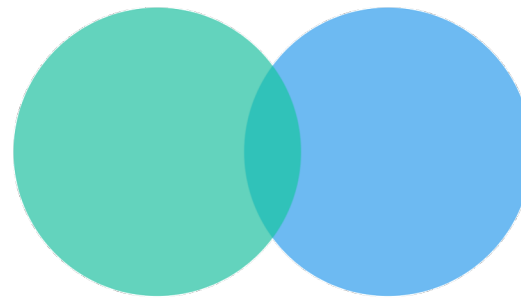
Measures



$p = .002$

Correlation: $r = .32$

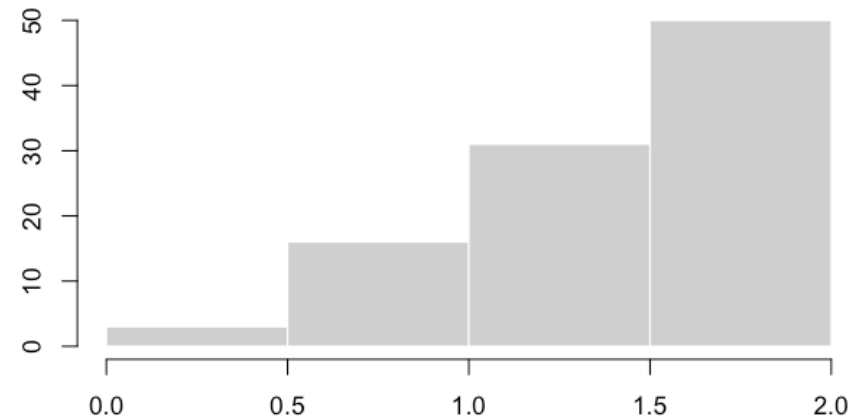
Shared Variance: 10%



Measures

● Language Skills

- 5-15 Questionnaire – Language Subscale
 - My child has difficulty understanding words.
 - My child has difficulty saying single words and short sentences.



- $M = 1.44, SD = 0.41$
- Range: 0 - 2

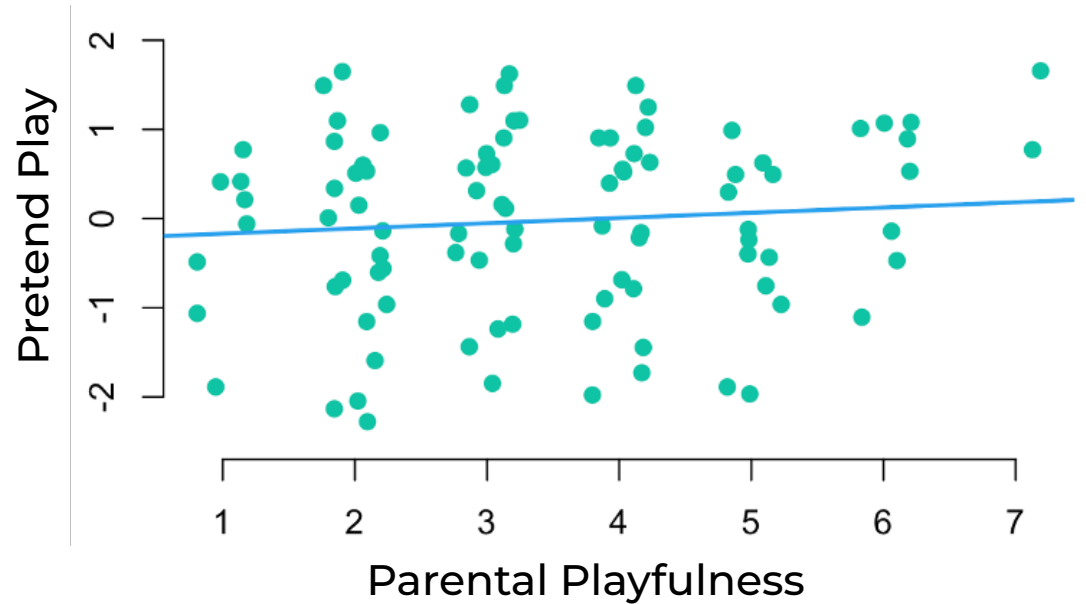


Findings



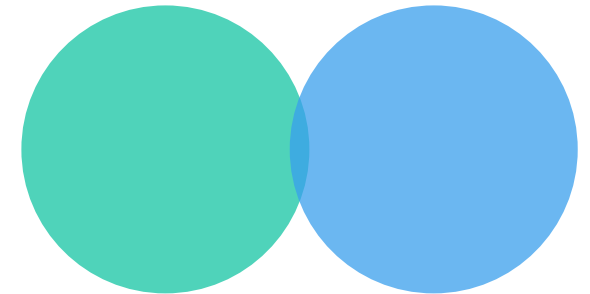
Pretend Play and Parental Playfulness

Two unrelated constructs?



Correlation: $r = .13$

Shared Variance: 2%

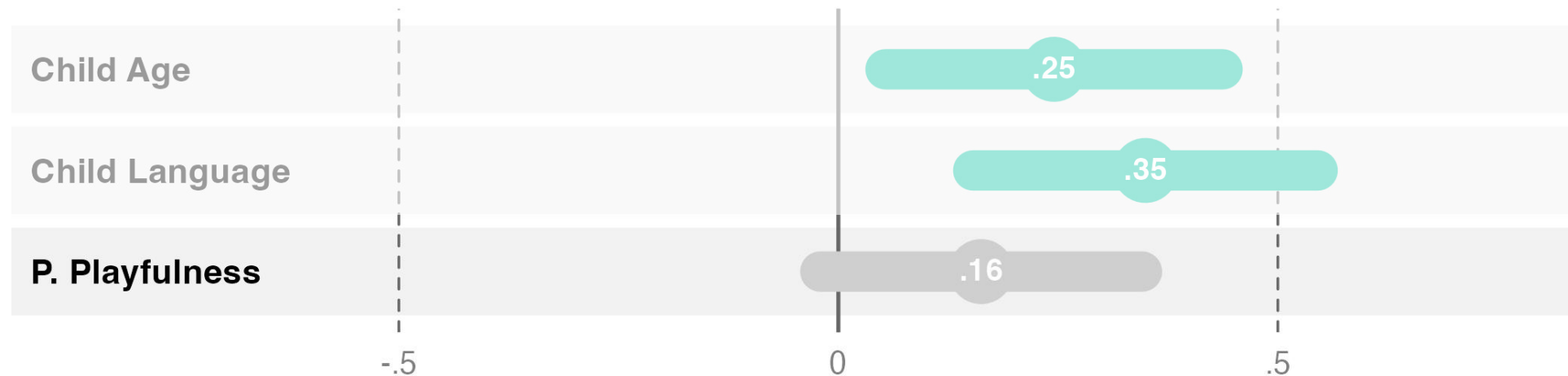


Pretend Play and Parental Playfulness

Two unrelated constructs?

Predicting Toddlers' Pretend Play Ability

$F(4, 92) = 10.38, p < .001, \text{Adj. } R^2 = .23$



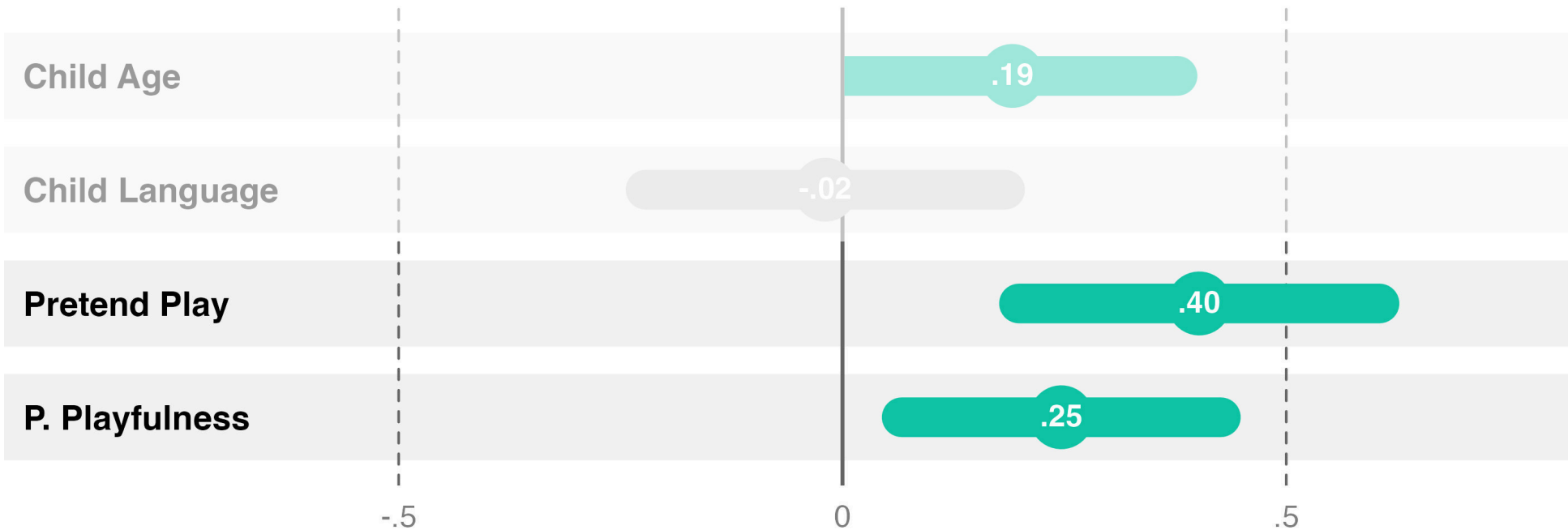
$N = 96$. Standardised coefficients and their 95% confidence interval are shown.

Play and Executive Functions

Unique contributions of parents and toddlers play

Predicting Toddlers' Working Memory

$F(4, 88) = 10.82, p < .001, \text{Adj. } R^2 = .30$



$N = 93$. Standardised coefficients and their 95% confidence interval are shown.

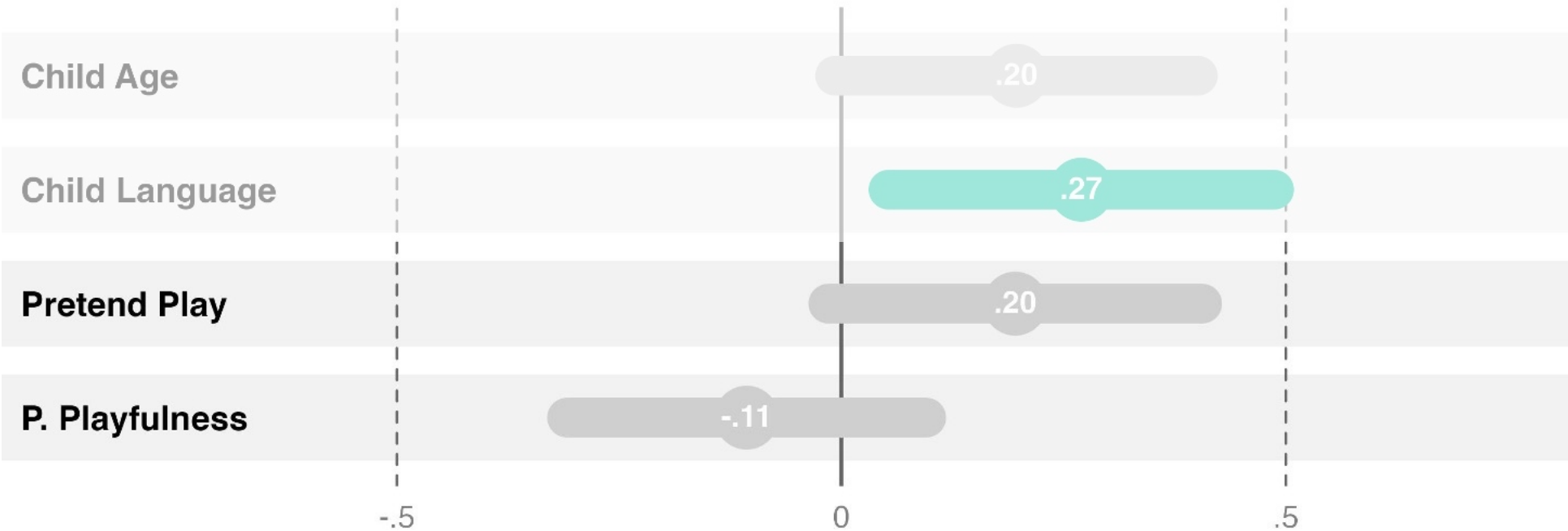


Play and Executive Functions

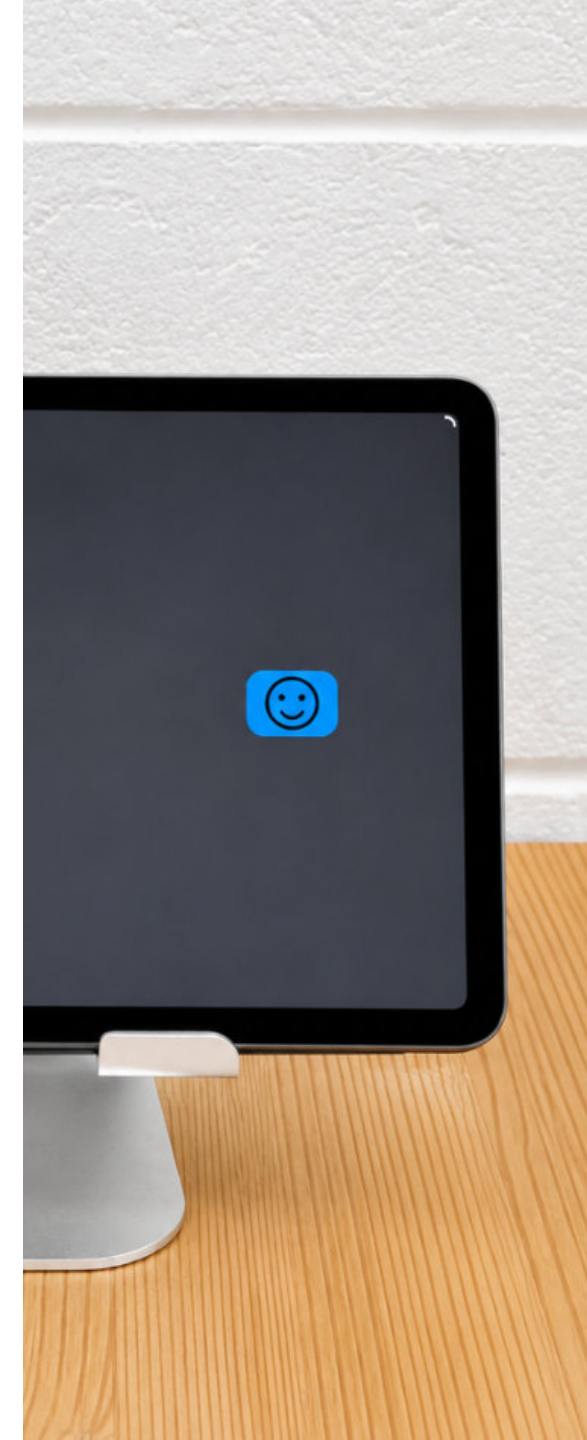
Unique contributions of parents and toddlers play

Predicting Toddlers' Response Inhibition

$F(4, 80) = 7.11, p < .001, \text{Adj. } R^2 = .23$



$N = 85$. Standardised coefficients and their 95% confidence interval are shown.



The Current Study

Hypothesis 1

Toddlers' ability to engage in pretend play positively predicts toddlers' executive functions.

→ *And so does parental playfulness.*

Hypothesis 2

Parental playfulness is not associated with toddlers' ability to engage in pretend play.

→ *At least not cross-sectionally.*



Take Aways

● Not *just* play

Executive functions are shaped through **interactions**, specifically in instances where children are required to **maintain representations**.

- Current findings reflect previous meta-analytic results and theoretical frameworks.
- Unique associations to working memory but not inhibitory control may indicate age effects.
- Longitudinal stability of relations requires further study.



Take Aways

● Being **silly** pays off

Parental playfulness has broader implications for children's development than shaping children's play.

- Playful interactions create manageable but challenging cognitive demands (i.e., *zone of proximal development*).
- Reflects previous findings regarding developmental benefit of parental playfulness.



Take Aways

- **Let's pretend!**

Children's pretend play and parental playfulness may be trained and stimulated.

- **Let's explore!**

Measures and conceptualisations of **adult playfulness** are not uniform. What does it mean to be playful?

- **Let's wait!**

Current findings are only cross-sectional. Further longitudinal research is needed.





Ways to be *playful*

- Build a tower with stuff you have lying around on your desk.
- Make up the cast of Harry Potter only using your friends.
- Challenge a friend to a round of Tic-Tac-Toe.
- Take a different way home.
- Write a two-line poem.

Thank you for your attention,
at this Psychology convention!

Stay in touch!

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*“It’s easy to fly: Just bend
your knees and jump!”*

(Child 20, cited in Ramsden et al. 2026, p. 12)

